

PROJECT DESIGN PHASE

Project Name: Import Data Using Transform Maps (Spreadsheet)

Module 2: Proposed Solution

1. Executive Summary

The proposed solution introduces an automated, scalable, and reliable data ingestion framework leveraging an **Import Set and Transform Map Engine**. This solution eliminates manual data entry errors by staging incoming spreadsheet files (.xlsx or .csv) in an intermediate layer before applying transformation rules to update target database tables accurately.

2. Core Features & System Capabilities

- **Automated File Parsing:** Uploads and parses standard Microsoft Excel (.xlsx) and CSV spreadsheets directly.
- **Intermediate Staging Layer (Import Set):** Isolates incoming raw data in a staging table to protect target production tables from corrupt records.
- **Flexible Mapping Engine:** Offers Auto-Map Matching for identical field names and Mapping Assist for custom column-to-field alignment.
- **Smart Deduplication (Coalesce Keys):** Uses unique identifier fields (e.g., Email, ID) to update existing database records instead of creating duplicate entries.
- **Audit Logging & Reporting:** Captures complete logs of inserted, updated, and errored records for every import run.

3. Proposed Workflow Steps

Step 1: User uploads raw Excel/CSV file → **Step 2:** System creates Import Set staging table → **Step 3:** Transform Map applies mapping & Coalesce checks → **Step 4:** Target table receives inserted/updated records with full audit logs.

4. Key Business Outcomes

- **Efficiency:** Eliminates manual keying overhead for bulk data loads.
- **Data Integrity:** Ensures 100% schema alignment and zero duplicate records.